Cross-Dataset Federated Learning for Floodwater Segmentation: A Literature Review and Proposed Comparative Study

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***Abstract*—Accurate floodwater segmentation supports emergency response, exposure assessment, and recovery planning, yet robust models require geographically diverse labeled imagery that is often distributed across separate data owners. Federated learning (FL) permits collaborative model training while raw images remain at their source, but strong cross-dataset heterogeneity can cause unstable optimization and client drift. This review analyzes 20 studies spanning federated optimization, federated semantic segmentation, and remote-sensing flood mapping. It motivates a controlled comparison of FedAvg, FedProx, FedNova, and SCAFFOLD using one independently collected Sentinel-1 flood dataset per client: ETCI 2021, MMFlood, and Kuro Siwo. A common U-Net and harmonized binary target will isolate the effect of the FL optimizer. The reviewed evidence shows that advanced algorithms address different types of heterogeneity and that no method is uniformly superior. It also shows that flood studies remain primarily centralized or single-dataset. The resulting gap is the absence of a controlled comparison of established FL optimizers for temporary-floodwater segmentation under real cross-dataset client heterogeneity.**

***Keywords*—federated learning, floodwater segmentation, non-IID data, Sentinel-1, U-Net, remote sensing**

I. Introduction

Floods are among the most destructive natural hazards, causing loss of life, displacement, infrastructure damage, and long-term economic disruption. Rapid delineation of inundated areas is therefore important for emergency response, resource allocation, exposure assessment, and post-disaster recovery. Earth-observation systems can survey large or inaccessible areas, while Synthetic Aperture Radar (SAR), particularly Sentinel-1, remains useful during flood events because it operates day and night and is less obstructed by clouds than optical sensing. Flood mapping is nevertheless not identical to general water detection. Permanent rivers, lakes, and reservoirs may already be present before an event; operational mapping must identify temporary inundation without treating all surface water as newly flooded [14], [17].

Semantic segmentation provides a pixel-level solution to this problem. Encoder–decoder networks such as U-Net learn both contextual and spatial information and have produced useful flood masks from Sentinel-1 imagery [13], [18]. Their reliability, however, depends on representative annotations across events, climates, land covers, and acquisition conditions. Public flood datasets differ in geographic coverage, sensor preprocessing, label generation, class balance, temporal context, and sample count [16], [17], [20]. A model trained on one source can consequently perform poorly when deployed on another domain; recent cross-benchmark evidence confirms that individual-dataset models frequently lose accuracy outside their training distribution [19].

Centralized multi-dataset training can increase diversity, but it assumes that every institution can transfer imagery and annotations to one location. Storage, bandwidth, ownership, governance, and geospatial-security constraints may make that assumption impractical. FL instead sends a shared model to data-holding clients, performs local optimization, and aggregates model updates at a server. Raw imagery remains local, although FL alone is not a formal privacy guarantee because protection against information leakage requires mechanisms such as secure aggregation or differential privacy.

The standard FedAvg algorithm aggregates client parameters in proportion to local sample counts [2]. Its simplicity and communication efficiency make it the principal baseline, but several independent flood datasets create more than an artificial label imbalance. Each client may have a distinct feature distribution, geography, annotation process, class ratio, and dataset size. Local updates can therefore move in conflicting directions. FedProx constrains local deviation from the global model [3]; SCAFFOLD uses control variates to correct client drift [4]; and FedNova normalizes updates when clients perform unequal amounts of local computation [5]. NIID-Bench demonstrates that their rankings change across heterogeneity settings, so superiority cannot be assumed without task-specific evaluation [6].

The base study by Islam applied a U-Net and FedAvg to 290 grayscale aerial flood images divided among three simulated clients [1]. Its federated model achieved 0.7186 IoU and 0.8363 F1, compared with 0.7450 IoU and 0.8539 F1 under centralized training; federated recall was higher (0.8883 versus 0.8638) but precision was lower (0.7900 versus 0.8443). These results establish feasibility, yet all clients were partitions of one Kaggle dataset. Their non-IID behavior came from

unequal sample counts and flood-pixel ratios, not independent geographic domains. The study also compared only FedAvg and did not report a separate numerical cross-dataset evaluation.

This project addresses that limitation by assigning ETCI 2021, MMFlood, and Kuro Siwo to three separate clients. Each source will be harmonized to Sentinel-1 VV/VH inputs and a binary temporary-floodwater target. A common U-Net, preprocessing policy, initialization, and training budget will be used so that the principal independent variable is the FL optimizer. The aim is to determine how FedAvg, FedProx, FedNova, and SCAFFOLD behave under genuine cross-dataset heterogeneity rather than a partition of one source. The objectives are to:

1) construct compatible client pipelines while preserving the separation of the three datasets;
2) compare the four algorithms using IoU, Dice/F1, precision, recall, convergence, and per-client performance;
3) benchmark federated models against centralized and local-only reference models;
4) examine whether global gains are distributed fairly across clients rather than dominated by the largest dataset; and
5) identify the optimization strategy most robust to real flood-domain heterogeneity.

The remainder of this review groups prior work into federated optimization, federated segmentation under domain shift, and flood-specific segmentation and datasets. It then derives the research gap, problem statement, and research questions that govern the proposed comparison.

II. Literature Review

This narrative review covers 20 studies selected for direct relevance to at least one of three components: heterogeneity-aware FL optimization, federated semantic segmentation, or flood segmentation and dataset generalization. Sixteen were published between 2021 and 2025. Four older works are retained because they introduce the four algorithms being compared. Tables I–IV report each study's method, dataset, evaluation measures, representative performance, and principal limitation. Scores are reproduced as reported and are not directly rankable because tasks, splits, architectures, and label definitions differ.

A. Federated Optimization Under Heterogeneity

FedAvg established iterative local training followed by weighted parameter averaging [2]. Its communication savings are substantial, but additional local epochs can amplify drift when client objectives differ. FedProx adds a proximal penalty around the current global parameters and is therefore relevant when independent flood clients produce divergent updates [3]. Its behavior depends on the coefficient μ: weak regularization resembles FedAvg, while excessive regularization suppresses useful local learning.

SCAFFOLD estimates server and client control variates to correct local directions [4]. Its strong results under severe label heterogeneity make it theoretically attractive for cross-dataset clients, although maintaining control states increases implementation and communication cost. FedNova addresses objective inconsistency by normalizing updates for unequal local work [5]. Consequently, its central advantage is not fully tested if all clients always execute identical local steps. A sound comparison should use equal computation for the main statistical-heterogeneity experiment and, if resources permit, add an unequal-local-step stress test.

NIID-Bench is especially important because it compares all four methods across several non-IID constructions and finds no universal winner [6]. However, these are mostly artificial partitions of classification datasets. Islam's flood experiment is closer to the target application, but its clients still originate from one aerial dataset [1]. Table I shows that optimization evidence is strong, whereas direct evidence for independent flood domains remains absent.

B. Federated Segmentation and Domain Shift

Federated semantic segmentation adds pixel imbalance and spatial consistency to the optimization problem. FedDG uses episodic learning and frequency-space interpolation to transfer distributional information among medical clients and improve unseen-domain performance [7]. FedDrive studies autonomous-driving segmentation under statistical and appearance heterogeneity; its results show that naive FedAvg can degrade sharply and that local normalization and style augmentation can recover performance [8]. These studies support evaluation on independent domains, but medical and RGB driving styles differ physically from SAR backscatter.

FedSeg addresses class-heterogeneous segmentation through a background-aware loss and pixel-level contrastive learning [9]. FedST uses diffusion-based style transfer to synthesize cross-domain appearances [10]. Their gains demonstrate that segmentation-specific interventions can exceed optimizer-only baselines, but they alter losses, representations, or data distributions. They are therefore related work rather than fair replacements for the proposed controlled optimizer comparison.

GeoFed is the closest remote-sensing precedent. It models coupled class-distribution and object-appearance heterogeneity across institutions and improves land-cover/building segmentation over FedAvg [11]. FedSaaS similarly aligns local and global class representations with prototypes, contrastive losses, and adversarial harmonization [12]. Both improve accuracy but introduce several components and, in FedSaaS, exemplar transfer. The proposed project deliberately keeps U-Net and the loss fixed to measure the behavior of four established optimizers before adding segmentation-specific mechanisms.

C. Flood Segmentation Models and Evaluation

Flood studies show that strong in-domain segmentation does not by itself establish cross-domain robustness. Paul and Ganju combine U-Net ensembles, pseudo-labeling, and conditional random-field post-processing on the ETCI challenge [13]. Pech-May et al. train U-Net on Sentinel-1 Sen1Floods11 and report high precision but lower recall, illustrating the risk of missed flood pixels despite favorable aggregate accuracy [18]. Bai et al. explicitly separate permanent and temporary water, and their lower temporary-water scores confirm that floodwater is harder than generic water segmentation [14]. This

TABLE I. LITERATURE SURVEY, PART I: FEDERATED OPTIMIZATION AND THE BASE FLOOD-FL STUDY.

Ref.	Year	Method	Dataset(s)	Metrics	Reported performance	Key limitation
Islam [1]	2025	U-Net with three-client FedAvg; centralized comparison	Kaggle Flood Area Segmentation; one dataset split into 116/54/62 client samples	Precision, recall, F1, IoU, MPA, FWIoU	Centralized/FedAvg IoU: 0.7450/0.7186; F1: 0.8539/0.8363; recall: 0.8638/0.8883	Artificial partitions of one aerial dataset; only FedAvg; no separate numeric cross-domain test
McMahan et al. [2]	2017	Weighted averaging after local SGD	MNIST, CIFAR-10, Shakespeare	Accuracy, rounds	MNIST target reached in 18 IID rounds versus 626 with FedSGD; 173 versus 483 non-IID rounds	Classification/language focus; susceptible to client drift under strong domain shift
Li et al. [3]	2020	Proximal local objective around global weights	Synthetic, MNIST, FEMNIST, Shakespeare, Sent140	Accuracy, convergence	About 22 percentage-point mean accuracy gain over FedAvg in a highly heterogeneous/straggler setting	Requires tuning μ; no dense segmentation evaluation
Karimireddy et al. [4]	2020	Server/client control variates correct update directions	EMNIST and synthetic tasks	Accuracy, rounds	Severe EMNIST setting reached target in 77 rounds versus 258 for FedAvg; 0.801 versus 0.787 accuracy at 1000 rounds	Extra state and communication; classification rather than segmentation
Wang et al. [5]	2020	Normalized aggregation for unequal local computation	Non-IID CIFAR-10, VGG-11	Accuracy	Random local epochs: 73.22% versus 64.22% for FedAvg; with momentum 77.07% versus 70.44%	Main benefit depends on variation in local work; no segmentation task
Li et al. [6]	2022	NIID-Bench comparison across six non-IID constructions	Nine classification/tabular datasets	Accuracy, convergence	One CIFAR-10 setting: SCAFFOLD 69.8%, FedAvg 68.2%, FedProx 67.9%, FedNova 66.8%; rankings change by setting	Artificial partitions and non-segmentation tasks

TABLE II. LITERATURE SURVEY, PART II: FEDERATED SEMANTIC SEGMENTATION AND DOMAIN GENERALIZATION.

Ref.	Year	Method	Dataset(s)	Metrics	Reported performance	Key limitation
Liu et al. [7]	2021	Episodic learning in continuous frequency space	Multi-centre fundus and prostate MRI	Dice, Hausdorff distance	Fundus Dice: 87.03% versus 85.01% FedAvg; HD: 13.07 versus 15.93	Medical domains; frequency/style assumptions may not represent SAR variation
Fantauzzo et al. [8]	2022	FL benchmark with local normalization and style transfer	Cityscapes, IDDA	mIoU	Heterogeneous Cityscapes: FedAvg 43.33%; SiloBN+LAB 53.37%; rainy unseen domain 53.99%	RGB driving scenes; different client scale and appearance physics
Miao et al. [9]	2023	Background-aware loss plus pixel contrastive learning	Cityscapes, CamVID, Pascal VOC, ADE20K	mIoU, pixel accuracy	Strong non-IID Cityscapes mIoU: 50.24% versus 10.40% FedAvg; CamVID: 63.50% versus 19.06%	Artificial class heterogeneity; changes the loss and representation mechanism
Ma et al. [10]	2024	Diffusion style transfer for cross-client image synthesis	Material, CHAOS medical, race and age segmentation	Dice, VI, MPA	Medical Dice: 0.9233 versus 0.3622 FedAvg; material Dice: 0.8145 versus 0.6754	High generation cost and additional information-sharing assumptions
Tan et al. [11]	2024	Global insight, essential features, and local-global balance	FedFBP, FedCASID, FedInria	mIoU/IoU	FedFBP: 66.70% versus 63.13% FedAvg; FedCASID: 74.34% versus 71.18%	Preprint; land-cover/building tasks; several coupled modules
Yu et al. [12]	2025	Global class prototypes and local adversarial harmonization	Driving-scene client domains	Accuracy, mIoU	Severe shift mIoU: 48.67% versus 37.19% FedAvg; unseen domain: 45.96%	Exemplar upload and high overhead; RGB driving rather than SAR

distinction is central to the proposed label policy: temporary floodwater is positive; permanent water and dry land are negative; uncertain or invalid pixels must be ignored rather than silently converted to background.

Portalés-Julià et al. provide the strongest direct evidence for the proposed gap by evaluating flood models across several Sentinel benchmarks [19]. Joint centralized training improves transfer, whereas single-dataset models struggle outside their source. Their work demonstrates the value of cross-dataset diversity but presumes that data can be pooled. The proposed study asks whether FL can obtain collaborative benefit while maintaining the three dataset silos.

D. Flood Datasets and Domain Compatibility

Dataset choice determines whether client heterogeneity is scientifically meaningful or merely incompatible. FloodNet supplies rich UAV imagery and multiple damage classes, but its RGB aerial modality and post-Hurricane-Harvey setting do not match a common Sentinel-1 VV/VH model input [15]. It is valuable literature evidence, not an appropriate client for the final homogeneous-channel experiment. DeepFlood similarly adds high-resolution aerial and SAR data plus inundated-vegetation labels, but its four-class regional design and mixed resolutions would introduce additional experimental factors [20].

MMFlood and Kuro Siwo are better aligned with satellite flood delineation. MMFlood covers 95 events in 42 countries and links Sentinel-1 acquisitions with CEMS flood maps and auxiliary terrain/hydrography data [16]. Kuro Siwo covers 43 events and explicitly represents floodwater, permanent water, and non-water using multi-temporal dual-polarization SAR [17]. Together with ETCI, these datasets provide independent acquisition and annotation pipelines while retaining a common Sentinel-1 representation. Their compatibility is not automatic: masks, no-data conventions, spatial resolution, normalization, and event-level splits must be audited before training.

E. Critical Synthesis, Research Gap, and Problem Statement

The reviewed literature exposes three recurring trade-offs. First, optimization methods target different failure modes. FedProx controls local deviation, SCAFFOLD corrects drift direction, and FedNova corrects unequal local computation. Their

TABLE III. LITERATURE SURVEY, PART III: FLOOD-SEGMENTATION METHODS AND CROSS-DATASET EVALUATION.

Ref.	Year	Method	Dataset(s)	Metrics	Reported performance	Key limitation
Paul and Ganju [13]	2021	Semi-supervised U-Net ensembles, pseudo-label filtering, CRF	ETCI/NASA Impact Flood Detection, Sentinel-1	IoU	IoU 0.7654 versus approximately 0.60 challenge baseline	Centralized, challenge-specific pipeline; depends on pseudo-label quality
Bai et al. [14]	2021	BASNet-style segmentation with focal, IoU, and SSIM losses; S1/S2 fusion	Sen1Floods11	mIoU, IoU, omission, commission, OA	Temporary-water test: 52.99% mIoU, 52.30% IoU, 92.81% OA	Optical availability/cloud sensitivity; temporary water remained difficult
Pech-May et al. [18]	2024	U-Net on dual-polarized Sentinel-1	Sen1Floods11	Accuracy, precision, recall, F1, IoU	At 200 epochs: precision 0.9431, recall 0.7639, F1 0.8441, IoU 0.7302	Single centralized dataset; event-level separation not established; SAR-shadow errors
Portalés-Julià et al. [19]	2025	Cross-dataset U-Net comparison across S1, S2, and multimodal inputs	Kuro Siwo, WorldFloods, S1S2-Water; tests on UNOSAT and Sen1Floods11	IoU	Multimodal model: 83.55 IoU points on UNOSAT and 75.97 on Sen1Floods11; joint training improved transfer	Centralized pooling; modality and label harmonization not studied under FL

TABLE IV. LITERATURE SURVEY, PART IV: FLOOD DATASETS AND BENCHMARK EVIDENCE.

Ref.	Year	Method/contribution	Dataset(s)	Metrics	Reported performance	Key limitation
Rahnemoonfar et al. [15]	2021	UAV post-flood dataset with segmentation, classification, and VQA baselines	FloodNet; 2343 high-resolution images, 10 segmentation classes	Per-class IoU, mIoU	PSPNet mIoU 79.69%; DeepLabV3+ 61.53%; ENet 42.61%	UAV RGB and one event; class semantics do not directly equal temporary floodwater
Montello et al. [16]	2022	Multimodal S1, DEM, hydrography, and CEMS delineations	MMFlood; 1748 acquisitions, 95 events, 42 countries	mIoU	Strongest reported binary benchmark approximately 66.52% mIoU	CEMS label inconsistency/noise; auxiliary modalities are not common to all clients
Bountos et al. [17]	2024	Multi-temporal VV/VH SAR benchmark with permanent-water separation	Kuro Siwo; 43 events across six continents	Class F1, mIoU	Three-acquisition U-Net-ResNet50: 76.20% mIoU; flood F1 80.12%; permanent-water F1 78.24%	Strongest setup uses temporal context; event and class imbalance
Fawakherji et al. [20]	2025	High-resolution aerial/SAR dataset for inundated vegetation	DeepFlood; four semantic classes	mIoU, class IoU, precision, recall	U-Net with high-resolution RGB+S1: 72.4% mIoU; inundated-vegetation IoU 77.7%	Regional four-class data; modality/resolution mismatch with S1-only clients

benefits should therefore be interpreted against the heterogeneity actually present rather than treated as interchangeable "improvements" over FedAvg. Second, federated segmentation papers achieve large gains by changing losses, normalization, prototypes, or style distributions. These methods demonstrate the importance of domain shift but confound a clean optimizer comparison. Third, flood papers use inconsistent meanings of water, floodwater, permanent water, and background. High overall accuracy can coexist with poor temporary-flood detection because background pixels dominate.

Evaluation must therefore include IoU and Dice/F1 as primary measures, supported by precision and recall. A pooled score alone is insufficient because the largest client may dominate it; macro-averaged and per-client results are needed. Event-level separation is also necessary to prevent neighboring patches from the same flood appearing in both training and testing. Because Kuro Siwo's strongest results use pre- and post-event imagery, restricting every client to post-event VV/VH improves input compatibility but removes useful change information. This trade-off must be disclosed rather than hidden.

The research gap follows from the intersection of the three bodies of work. Flood segmentation is mainly centralized or single-dataset [13], [14], [18], [19]; FL segmentation primarily studies medical, driving, or land-cover domains [7], [8], [11]; and the base flood-FL study partitions one aerial dataset and evaluates only FedAvg [1]. No reviewed study performs a controlled comparison of FedAvg, FedProx, FedNova, and SCAFFOLD for temporary-floodwater segmentation when each client owns an independently collected Sentinel-1 dataset.

The resulting problem statement is: *a U-Net trained through conventional FedAvg may converge poorly or favor particular clients when ETCI 2021, MMFlood, and Kuro Siwo create simultaneous geographic, feature, quantity, and annotation heterogeneity. It is unknown whether proximal regularization, normalized aggregation, or control-variate correction provides a more accurate, stable, and balanced global floodwater model under this realistic cross-dataset setting.*

The study is guided by four research questions:

1) How do the four algorithms compare in IoU, Dice/F1, precision, and recall under independent dataset domains?
2) Which method converges most stably and reaches useful validation performance in the fewest communication rounds?
3) Does the algorithm with the best pooled score also provide the strongest macro-average and per-client performance?
4) How closely can each federated model approach centralized training while improving over local-only models?

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